

ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	2799294
ToLID	qdGloPulc1
Species	Glomeris pulchra
Class	Diplopoda
Order	Glomerida

Genome Traits	Expected	Observed
Haploid size (bp)	438,212,396	361,559,850
Haploid Number	12 (source: ancestor)	13
Ploidy	2 (source: ancestor)	2
Sample Sex	Unknown	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for hap1: 7.7.Q56

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

- . Observed Haploid Number is different from Expected
- . Kmer completeness value is less than 90 for hap1
- . Assembly length loss > 3% for hap1
- . Not 90% of assembly in chromosomes for hap1
- . Assembly length loss > 3% for hap2
- . Not 90% of assembly in chromosomes for hap2

Curator notes

- . Interventions/Gb: 29
- . Contamination notes: "Repeated attempts to run BlobTools were unsuccessful, so we used an alternative tool called FCS-GX (Foreign Contamination Screen _ Genome eXclusion), to identify contaminated scaffolds. In this species, the -alt- includes 85 contaminated scaffolds with diverse bacterial sequences. Of these, four scaffolds exceeded 1 Mbp. The -prim- assembly contains 3 contaminated scaffolds. All contaminated scaffolds from both haps were flagged as contaminants during a preliminary curation step using pretextview, and were subsequently removed from the analysis in the pretextview file shared for review."
- . Other observations: "We generated a new diploid assembly of Glomeris pulchra from scratch. During initial trials, hifiiasm failed to correctly identify the heterozygous and homozygous coverage peaks, producing assemblies with an inflated number of duplicated BUSCOs. To stabilize the assembly graph, we re-ran hifiiasm with a manually specified homozygous coverage value (hom_cov = 86), matching the homozygous peak

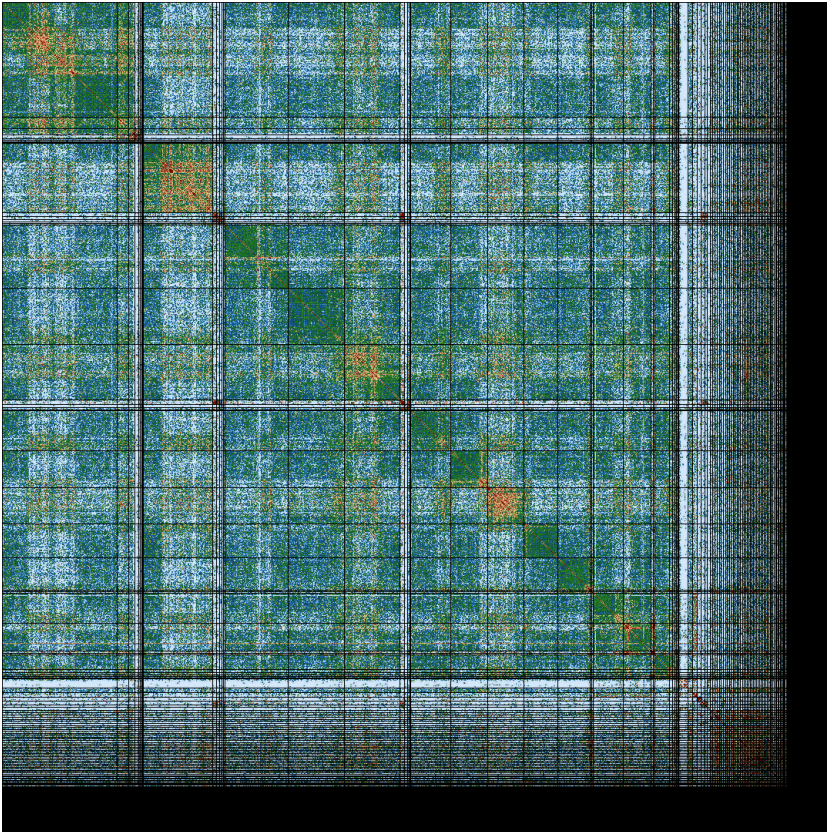
predicted by GenomeScope2 for the ONT read set. This adjustment substantially improved the assembly: duplicated BUSCOs were nearly eliminated, and no additional purge_dups step was required. We then curated the assembly in diploid mode. Because the Hi-C dataset was of relatively poor quality: approximately 68% of Hi-C reads did not map to the assembly. And because one haplotype (hap1) was noticeably more fragmented than the other, we prioritized optimizing the more contiguous haplotype (hap2). Several contigs originally assigned to hap1 were reassigned to hap2 when their structure and Hi-C signal supported this, and hap2 was designated as the primary assembly (pri or hap1 for compatibility with the EAR nomenclature). Hap1 was retained as the alternative assembly (alt or hap2). After curation, both the primary and alternative assemblies showed substantial improvements in contiguity, reflected in higher cN50 and sN50 values. Merqury QV scores also increased for both haplotypes (due to a bug in the EAR generating script only the QV of the prim/hap1 is shown at the report table), indicating improved consensus accuracy. The final Pretext maps for each assembly underwent two rounds of internal review, and save_state files have been provided for both assemblies. Finally, to evaluate whether the two *G. pulchra* samples (GP7 and GP16) belonged to the same species and to identify the most suitable assembly for downstream curation, we compared them using HoBRAC and Mash. HoBRAC performs reference-free structural validation by combining whole-genome alignment, BUSCO-anchored synteny, and reference ranking to detect misassemblies, large-scale rearrangements, and haplotig artifacts, while Mash estimates genomic relatedness by rapidly comparing the k-mer sketches of two assemblies to compute an approximate evolutionary distance. Synteny analysis in HoBRAC showed strong chromosome-level collinearity between GP7 and GP16, and Mash reported a very small inter-assembly distance (0.0119), consistent with two individuals from the same species. BUSCO profiles were similarly high for both assemblies, with GP7 showing fewer duplicated BUSCOs and GP16 exhibiting a larger assembly size due to increased repeat content. Because the Hi-C library mapped poorly to GP16 (78% unmapped reads), only GP7 was curated further, and GP16 was retained as an uncurated but biologically validated conspecific assembly."

Quality metrics table

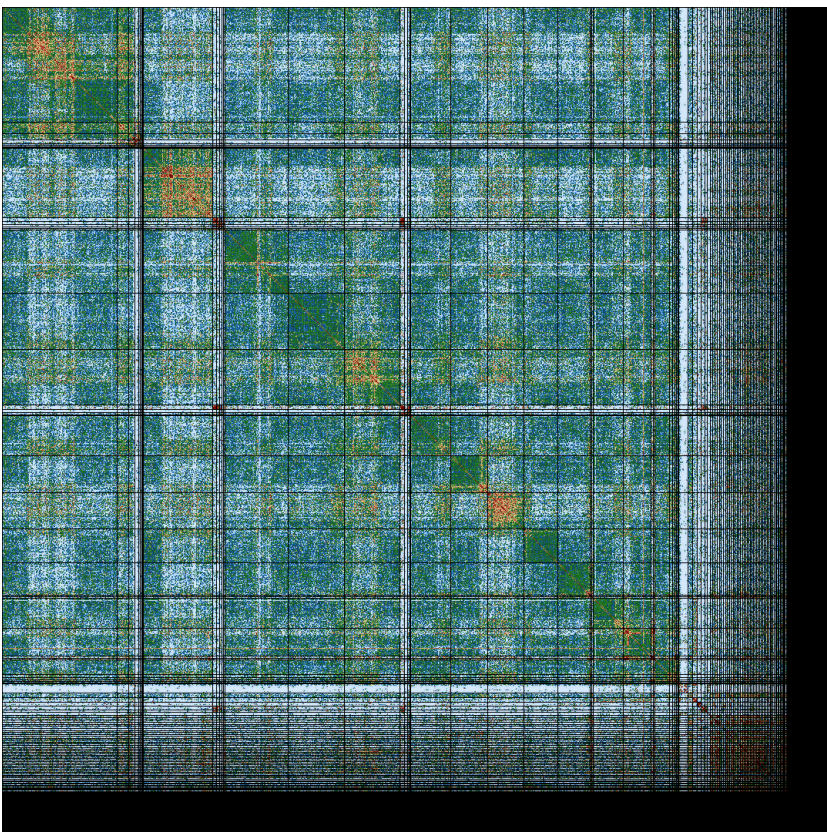
Metrics	Pre-curation hap1	Pre-curation hap2	Curated hap1	Curated hap2
Total bp	392,759,916	1,104,426,000	361,559,850	351,028,581
GC %	41.06	44.14	39.41	39.73
Gaps/Gbp	415.01	612.99	141.06	247.84
Total gap bp	32,600	135,400	10,200	17,400
Scaffolds	631	5,795	101	260
Scaffold N50	14,327,784	319,200	23,890,722	15,660,490
Scaffold L50	10	454	6	7
Scaffold L90	226	3,479	20	65
Contigs	794	6,472	152	347
Contig N50	8,886,200	276,670	10,975,611	8,123,070
Contig L50	16	513	11	14
Contig L90	301	4,076	40	96
QV	45.6609	50.3255	56.7606	
Kmer compl.	87.5619	78.0536	82.4639	
BUSCO sing.	97.5%	96.6%	96.5%	97.5%
BUSCO dupl.	0.6%	1.1%	0.6%	0.4%
BUSCO frag.	0.2%	0.3%	0.3%	0.2%
BUSCO miss.	1.7%	2.0%	2.6%	1.9%

BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: arthropoda_odb10 (genomes:90, BUSCOs:1013)

HiC contact map of curated assembly

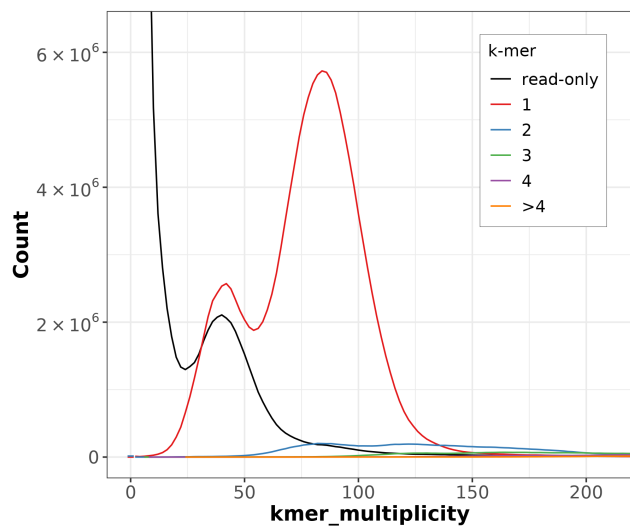


hap1 [\[LINK\]](#)

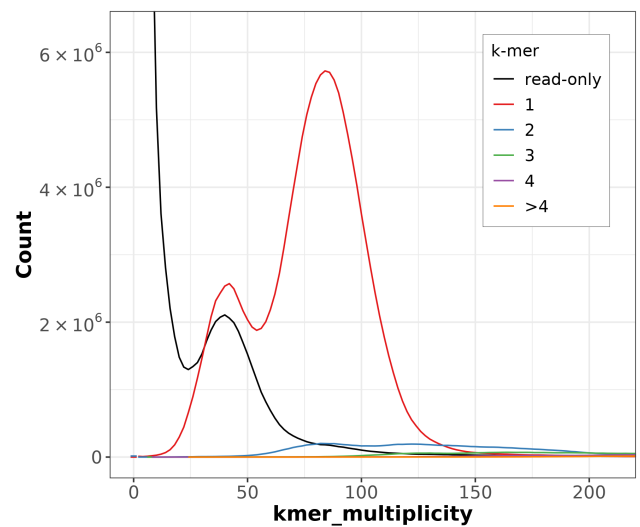


hap2 [\[LINK\]](#)

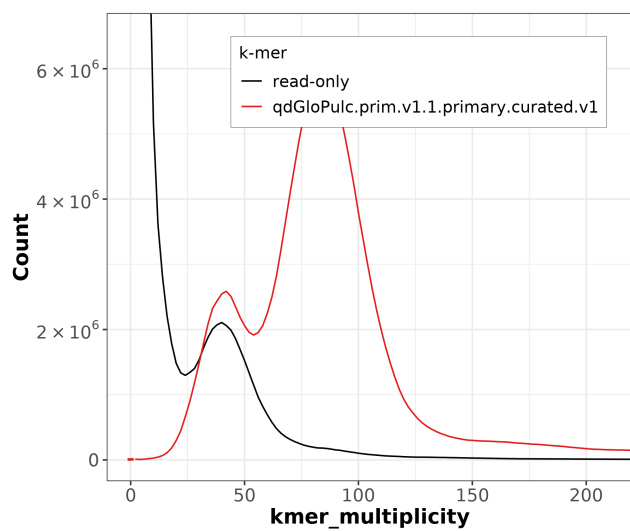
K-mer spectra of curated assembly



Distribution of k-mer counts per copy numbers found in asm

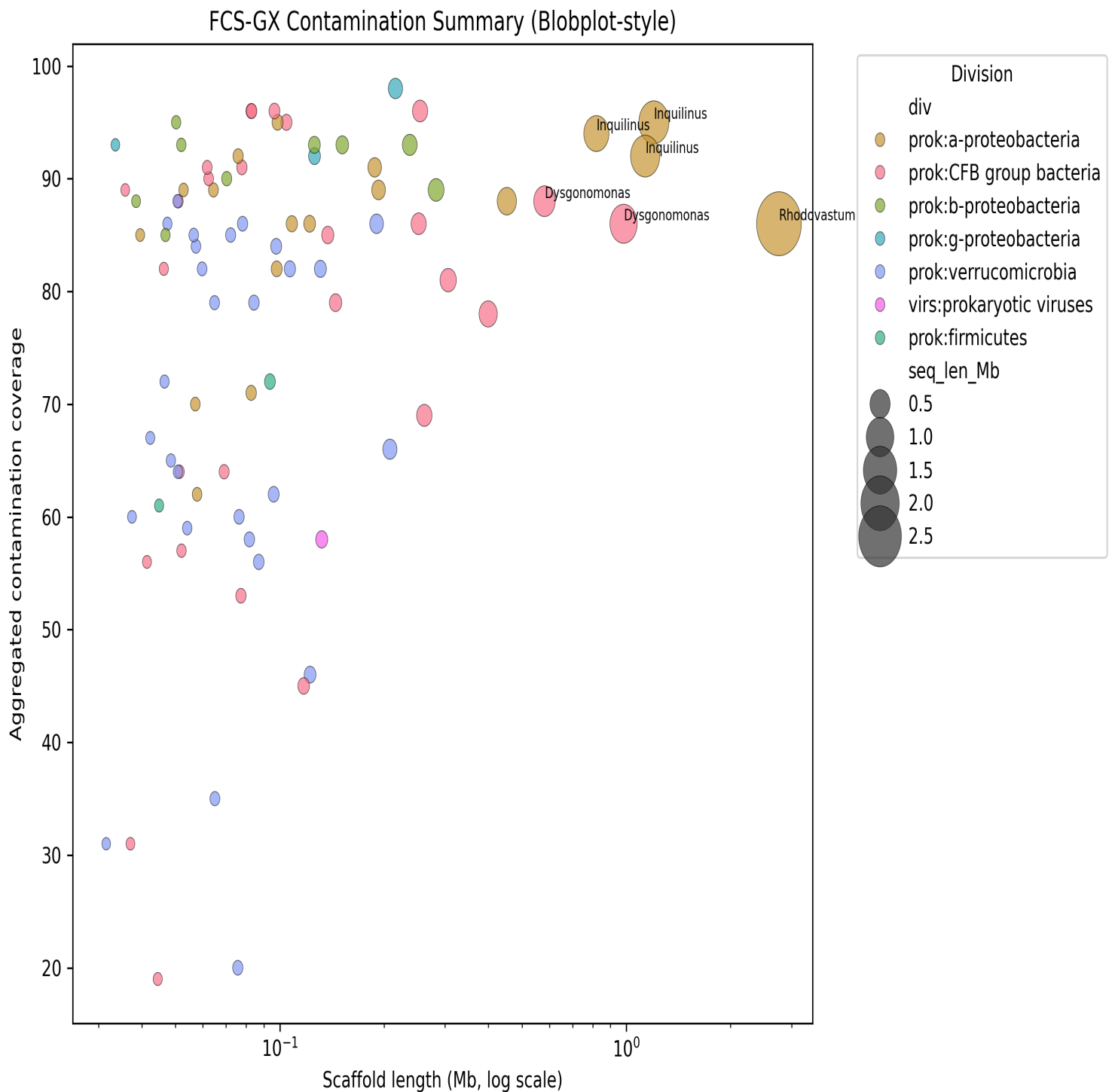


Distribution of k-mer counts per copy numbers found in asm



Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



hap1. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	ONT	Omni-C
Coverage	72	127

Assembly pipeline

- **CLAWS**
 - |_ *ver*: v3.1.0
 - |_ *key param*: NA
- **Filtlong**
 - |_ *ver*: v0.2.1
 - |_ *key param*: -t target_bases 40000000000
 - |_ *key param*: --filtlong-min-mean-q 99
- **Hifiasm**
 - |_ *ver*: 0.24.0
 - |_ *key param*: --ont
 - |_ *key param*: --hom-cov 86
- **Yahs**
 - |_ *ver*: 1.2a
 - |_ *key param*: NA

Curation pipeline

- **FCS-GX (ForeignContamination Screen-Genome eXclusion) pipeline**
 - |_ *ver*: v0.5.5-0
 - |_ *key param*: NA
- **PretextViewAI**
 - |_ *ver*: 1.0.7
 - |_ *key param*: NA

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